

Office of Chief Digital and Artificial Intelligence Officer's

Moving to a Canonical Controlled Vocabulary Model

“Data elements are aligned into a comprehensive data dictionary with a controlled, yet flexible, vocabulary and taxonomy.” – 2020 DoD Data Strategy

EXECUTIVE SUMMARY

The third goal of Department of Defense's (DoD) Data Strategy is to Make Data Understandable, with a key indicator of progress being the implementation of vocabulary that empowers the enterprise to “aggregate, compare, and truly understand data”. We must be able to manage our valuable digital assets as they transit across the enterprise, having confidence in understanding their intent and meaning based on the origin (provenance) and each context (Domain) that has been verified as valid for use. The vocabulary (data terms) must still support the tenets of visible, accessible, understandable, linkable, trustworthy, interoperable and secure (VAULTIS). To achieve this state of maturity in the data ecosystem, all data objects require the formality of a controlled vocabulary, local if isolated, and canonical if enterprise-wide, to ensure:

- Understanding
- Trustworthiness
- Interoperability

Going forward, each data term will be *tagged* so that use across the DoD will be:

- Unique and consistent across the DoD enterprise, by publishing vocabulary terms and definitions
- Persist for all time, inactivating versus deleting vocabulary terms allows for certainty in analytics
- Mapped to Labels, to allow natural interoperability between dissimilar languages
- Support most legacy systems, Current Labels in use can be maintained and mapped to Tags
- Human- and machine-readable

Progress is demonstrated when priority data products use is accelerated and specific data items or objects are consumed by multiple organizations without investment into translation or mediation services. This plan assumes a rapid pace for assessing current needs and capabilities, a regular drumbeat of agile development, and measured progress towards enterprise scale with specific goals for mission outcomes.

The CDAO will pursue achieving the new approach by:

- Prototyping capability – ‘learn by doing’ – for a controlled vocabulary model, pattern and service,
- Providing guidance on creation of controlled vocabulary terms and objects using patterns and processes for piloting and future incorporation into DoD Issuances,
- Determining if there is a DoD issuance that can be supplemented, or a new issuance is required
- Engaging with major programs, data producers and data consumers to drive enterprise adoption
- CDAO's initial steps to establish a controlled vocabulary implementation includes:
 - Staffing the Controlled Vocabulary model for community awareness and leveraging the JADC2 LOE 1 Task 1.2a as the foundational source material
 - Assessing the impact to major cross-Component UID programs (Figure 1)
 - Identifying early adopters, selected based on priority enterprise use cases, to pilot the approach
 - Issuing guidance formally establishing the approach and directing planning to adjust existing implementations

BACKGROUND

The Department of Defense (DoD) has embarked on a journey to digitally modernize operations.

“The Department of Defense's (DoD) mission is to provide the military forces needed to deter war and ensure our nation's security. The DoD now recognizes that data is a strategic asset that must be operationalized in order to provide a lethal and effective Joint Force that, combined with our network of allies and partners, sustains American influence and advances shared security and prosperity.” – 2020 DoD Data Strategy

The importance of data can no longer be overlooked as it is directly called out in the opening paragraph of the DoD Data Strategy Forward, and more practically, data and metadata are truly at the center of all things digital.

Enterprise Data Management (EDM) encompasses many aspects of data and the management of these data objects using diverse cooperative interoperated capabilities that are required to orchestrate the environment. This paper will focus on one of these concepts known as controlled vocabulary.

Cause: A lack of consistent data terms that are universally available.

The need to have a controlled vocabulary is driven by nature and diversity of DoD operations and partners. The concept of a data term is necessary for almost all digital processes and functions and permeates the entire enterprise. When terms are not immediately and unerringly understood by every enterprise participant, there is a significant risk that process outcomes and analytic results will be inconsistent between communities. The lack of consistency and comprehension of each data term also significantly impacts the ability to aggregate, consolidate interoperate information in a meaningful way.

Impact: Lack of understanding of specific data meaning and intent

The basic foundation for global understanding of data includes two fundamental items, a globally unique representation of a data term (the Tag) and the definition of the data term. Lacking this foundational information impacts the enterprise in many ways, including:

- Term Tag collision or deviation
 - Collision is when two distinct Tags both have the same definition
 - Deviation is when one Tag has two distinct meanings or definitions
- “Non-atomic” definitions – where there are two or more concepts embedded into the single definition

Impact: Inability to have independent domains create consistent outcomes and analytic results

There is no preventative measure to keep a single label from having two distinct definitions. It is unreasonable to expect the computers or humans to mediate the differences in all circumstances, certainly at the scale and speed the Department requires to fight the fight tonight.

Impact: Confidence in decision making is not achieved across the enterprise

If the terms collide there is significant risk in any decision-making based on the analytic results.

Challenge: inability to communicate with clear understanding and consistency at Scale

An environment without proper governance of controlled vocabulary creates challenges regarding explicit meaning and comprehension. This challenge has been identified many times in history, including the Hubble Space Telescope, where the units of measure in the specification for grinding the reflecting mirrors were not established using a controlled vocabulary, resulting in a repair mission. This is not tolerable when fighting the fight tonight.

TECHNICAL APPROACH AND CONSIDERATIONS:

Controlled Vocabulary Composition

Controlled Vocabulary is a simple concept at its core. It minimally only requires a symbolic term (the vocabulary item called a Data Tag) and a definition. However, practically speaking three additional elements make it extremely well suited for a data ecosystem that can support extremely large scale and distributed

operations. The five elements of a controlled vocabulary (Table 1) establish the pattern used to capture the full description of a controlled vocabulary entry.

Core	Term	Description
Yes	Data Tag (Term)	The symbolic term that is machine-readable and a globally unique identifier. This is persistent over time and across all communities
Yes	Definition	A machine-readable atomic (one concept) description or specification
No	UID	The machine-readable (and typically machine generated) unique identifier
No	Context	The domain of relevance (the functional area or mission space) that the Data Tag has been validated for relevance and use
No	Context Description	the human focused description of the Context to inform the knowledge workers of the composition and focus of the context

Table 1 - Elements of a Controlled Vocabulary

Each element of the controlled vocabulary (Top of Figure 1) has characteristics indicating machine and human readability, machine and human generation and allowed values for each instance of a controlled vocabulary term. These allowed values are likely to change from one term to the next.

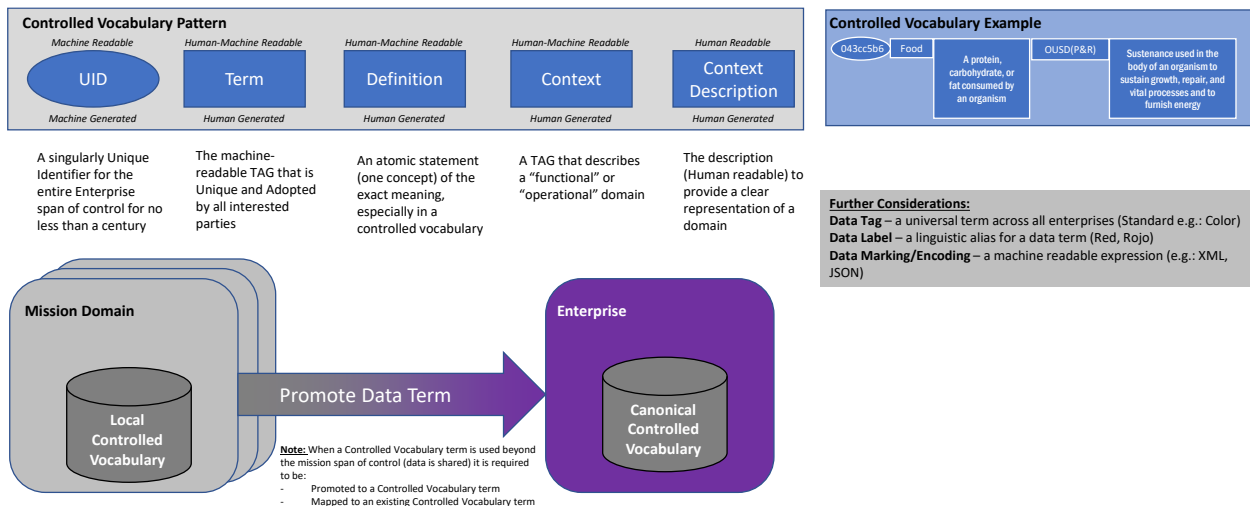


Figure 1 - Elements of a Controlled Vocabulary instance

Canonical versus Local Controlled Vocabulary (CCV/LCV)

One of the first considerations for any data is to identify when it will be canonical (enterprise-wide) versus local controlled vocabulary. If the data will be shared beyond the immediate mission space, then it is a canonical controlled vocabulary and must be promoted to enable enterprise sharing (Bottom of Figure 1). All controlled vocabulary must comply with the DoD Data Strategy and other guidance that promote interoperability across the DoD and beyond, as appropriate.

If the content will remain strictly local to the mission space, then an argument can be presented that the data itself is local. Given the current state of unknown analytic futures, this argument becomes very difficult to support. However, there are certain data that may be naturally local in nature where the local disposition of data is defensible (e.g: packet rate of exchange for a router in an isolated lab for a specific experiment).

Given situations where the data will never be exposed or shared with any “external” community, it is acceptable to manage the data as LCV. (Note: External in this case is any organizational entity that does not fall under the explicit governance, laws, regulations, policy, guidance, instructions, standards, patterns and practices of the local community. *We are not they*, and our rules, constraints and practices do not strictly apply.) However, in all other cases, it is prudent to simply assume the data will be CCV and prepare the

controlled vocabulary terms to be promoted, or a more efficient approach is to adopt and existing CCV and avoid the promotion process.

Proposed Differentiation Between Data Tags and Data Labels

Industry does not appear to have a clear distinction between a data tag and a data label. More often than not, tag and label are used interchangeably when discussing data. This situation can cause confusion, reduce understanding, or worse yet manifest distinct meanings for the “same vocabulary term” as the individuals are having the conversation, and have no idea they are communicating past one another.

However, if these terms are positioned with a distinct intent, then the unambiguous use of tag and label provides improved communication by establishing clear and concise understanding (Middle-Right Box of Figure 1). This approach also aligns with the controlled vocabulary premise that each term should have a distinct and unique atomic definition that has no conflict or overlap with other terms (We eat our own dog food).

The proposed model is straight forward and uses moderately traditional computer science concepts for alignment to create the distinct meanings for tag and label. The following constraints are applied:

- Data Term Tags are unique across the enterprise and have sustainable persistence over time
- Data Term Labels (lingual expression of a Tag [Red, Rojo]) are mapped to Tags
- Tags and Labels are further constrained as follows:
 - Tag – is a universal term that is language independent and always has the same meaning and definition, regardless of environment or language (This is an area where National Information Exchange Model (NIEM) is attempting to create a world-wide controlled vocabulary, however caution must be used prior to an outright acceptance of the NIEM offerings).
 - Label – is a lingual representation of a specific Tag and should only be mapped 1-1 to a Tag within a mission area. The Label is essentially an alias for a Tag. (Note: the use of “should” not “shall”. This is to accommodate the potential that a local label is used that also appears in other vocabularies with a different tag. This is not an issue as the information exchange and the mediation targets are always based on the Tag, not the Label. The Tag collision and deviation are managed in a de-confliction process described later in this paper. However, as the environment matures, labels will naturally homogenize over time.)

Behaviors and Constraints of Controlled Vocabulary

The proposed controlled vocabulary model ensures each data tag is machine-readable and is unique across the entire enterprise. This is necessary to assure a clear understanding of each specific term regardless of the originating mission or organization. Furthermore, it is necessary to know if the vocabulary term has been validated for use in a specific context or domain. Not all valid controlled vocabulary will be relevant or suitable for every context or domain. However, the term will be trustworthy and accurate, even if misapplied in a context or domain.

A basic flow diagram (Figure 2) is presented to identify the major activates that are envisioned to manage both the LCV and CCV, and the interoperation that must exist to make enterprise data more understandable and promote interoperability. The specific design and implementation details are not addressed in this discussion.

Logical Flow Diagram – New Vocabulary Term Request

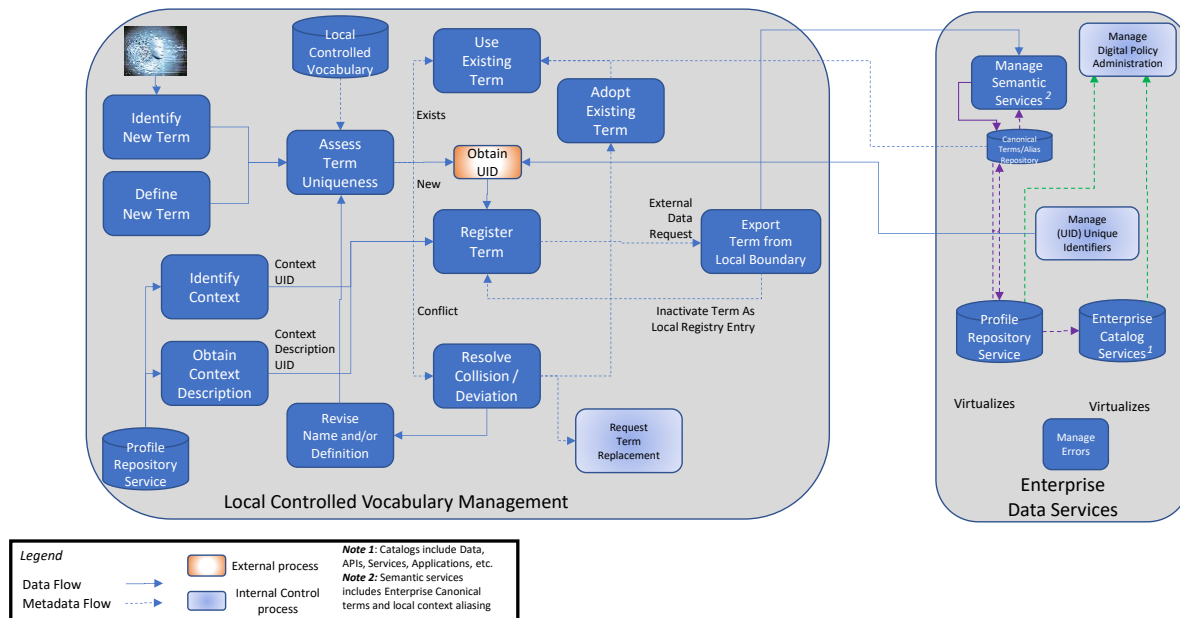


Figure 2- Basic Flow Diagram for Managing Controlled Vocabulary

The envisioned management process to define, capture, store and manage LCVs should be nearly identical, if not identical, to the enterprise process for CCVs. The basic process to create LCV terms applies to every new vocabulary term that is created. Even at the enterprise level, the enterprise is a local actor when crafting new terms, and then it will likely promote the term immediately to canonical controlled vocabulary. This nuance is important and must be managed as not all Office of the Secretary of Defense (OSD) or “Big DoD” data is actually enterprise data.

The key difference between a LCV and a CCV term is access and exposure. All CCV terms are fully accessible, given proper authorization, to the entire enterprise. LCV terms will likely be limited to the direct mission actors and not accessible or exposed beyond a closely controlled boundary.

Another key aspect of a controlled vocabulary is the ability to discretely identify each term uniquely. This is facilitated by assigning a UID to each LCV and CCV term to allow for full lineage tracking. This approach has two significant benefits:

- Every controlled vocabulary term can identify its Provenance (who created the term) – with the exception of legacy data where a data steward or an office of record will be “assigned” as the origination source to “help” historical record keeping. The ability to understand the original intended use of the data provides specific insight to help data consumers determine if it is suitable for individual contexts and domains. (This assumes the UID model has an organizational taxonomical issuing pattern that can be traced.)
- The application (use) of controlled vocabulary can be traced to create a full lineage record down to an individual data item. This has direct relevance to the financial audit processes and may prove very useful for other areas where knowing what exact data was used to generate a result becomes critical information.

Align and De-Conflict Controlled Vocabulary

A fully distributed independent cooperative service that manages the LCVs will eventually result in the duplication of a local controlled vocabulary term as it is promoted into the enterprise. To maintain the premise

of the CCV model, this requires an additional process to rectify collisions and deviations for any term that is promoted to a broader community or enterprise registration.

The concept of a **collision** results when two unique terms have the same fundamental definition. It is not required that the definitions be identical, but rather the interpretation of definitions results in the same understanding by one or more consumers.

The concept of a **deviation** is when one term has two fundamentally unique definitions. In this case, each definition's interpretation may result in unique understanding by one or more consumers.

The management of these two conditions must correct both the collision and deviations to preserve the uniqueness of the term across the enterprise. A potential alignment and de-confliction processes has been presented (Figure 3) to manage both collisions and deviations.

Logical Flow Diagram – Align/De-Conflict Vocabulary Term

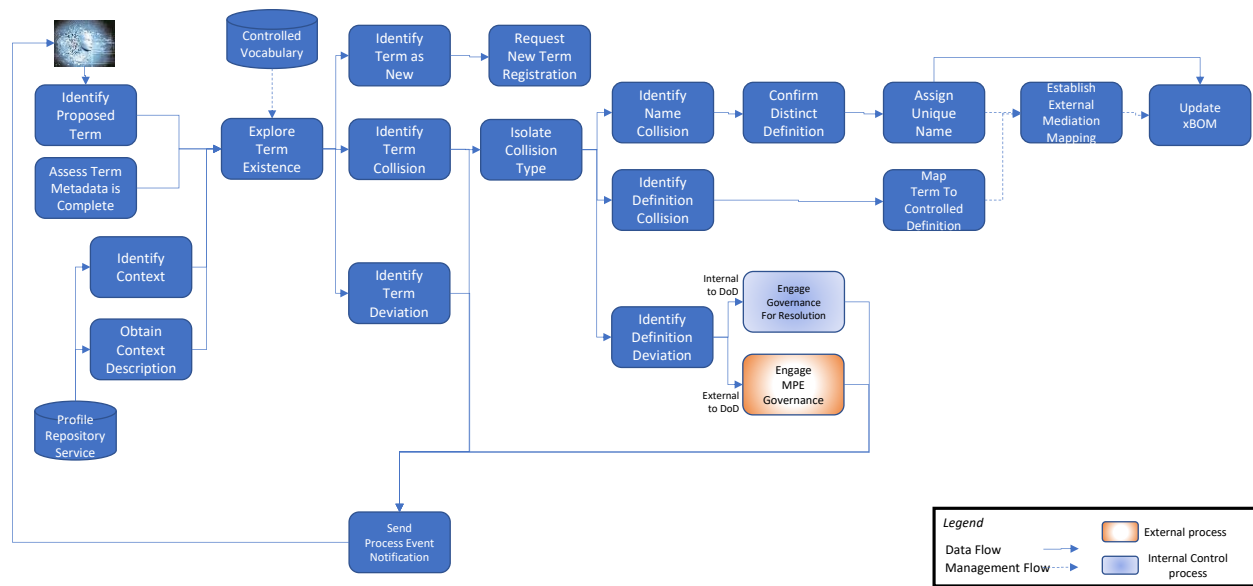


Figure 3 - Alignment and De-confliction process

Managing Tag Uniqueness of Controlled Vocabulary

The management of CCV terms will require a governance process.

There are basic process tenets that need to be achieved but are beyond the scope of this document to describe the details. The key outcomes should include, but may not be limited to:

- An DoD-wide empowered voting body of data experts that are not driven by fiscal or mission needs
 - Arbitration of term collisions
 - Arbitration of term deviations
 - Voting on promotion of terms into the Canonical Controlled Vocabulary
- A digital communications and workflow platform to accelerate decision and outcome timelines
 - Remote body assembly
 - Posting of pending actions for consideration and decision
 - Publishing of outcomes
 - Population of approved CCV Terms

- Agile practices to allow near-immediate decision making for registration of candidate CCVs
 - Workflow driven submissions and approvals
 - Automated notification and postings
 - A near real-time process to notify a rejected LCV petitioning for CCV status
- Approval of patterns that enable and promote maximum flexibility and innovation
- Issuance of standards that do not interfere with pattern application
- A formal process model (with time-boxing) to allow the enterprise to manage expectations

FUTURE TECHNICAL CONSIDERATIONS AND NEEDS:

This section identifies some of the known technology enablers that are ultimately required to implement and optimize the overall management of the CCV process. This section will not define or propose implementation or deployment approaches.

- A Tag generator that is used to instantiate new CCV (and LCV) terms – which allows the enterprise to create globally unique Tags for terms
 - This should be a replicable deployable micro/service that is controlled at the enterprise level and implemented in the field as most appropriate (necessary to support LCV and CCV independently but maintain strict alignment).
 - Generate the Controlled Vocabulary Tag Term -- This service confirms Tag uniqueness
 - Enterprise
 - Local (independent from enterprise, but could have an alignment feature to allow for collision/deviation avoidance and ghost registration to accelerate future promotion of LCV to CCV terms)
 - Map a Label to a Tag and confirm Tag-Label uniqueness within a context
 - Publish the Controlled Vocabulary Tag (including all known labels)
 - Map emergent Label to Tag within a Context (Domain) (future Label instantiations)
- Service to validate atomic definition
- Service to validate machine-readable/human-readable compliance
- Collision-Divergence Management Service
 - Identifies collisions
 - Conflict is two terms with the same definition
 - Divergence is the same term with two definitions
- Provides metrics on provenance and pedigree for Tags
- Mediation Services Management
 - Identifies existing mediation services (registry of mediation hubs/translation APIs)
 - Recommends resolution options for mediated Tags (and Labels indirectly)
 - Stores mapping patterns to accelerate future Tag-Tag or Tag-Label mediation
 - Ability to consume ontological yield maps
 - CCV Tags between domains
 - Dynamic inter-mediation enabler
- Enterprise hosting services
- Enterprise Software Factory
 - Includes maintenance of CCV suite services
 - Includes distribution services for CCV suite services distributed deployment
- Revision & Recall to allow longitudinal understanding

ENTERPRISE CAPABILITY:

The enterprise benefit of a managed canonical (and local) controlled vocabulary capability provides Visibility, Accessibility, Understanding and Trustworthy, Interoperability of VAULTIS as follows:

- Visibility– is achieved with the enterprise repository of all canonical controlled vocabulary search
- Accessibility – is achieved with an interface (User Interface/Application Programming Interface (UI/API)) to allow machine-to-machine (M-M) and Human-to-Machine (H-M) to obtain Term metadata
- Understandability – is achieved with unique Tags and associated atomic definitions of a vocabulary term and associated labels per context
- Trustworthy – is achieved by having computable confidence levels for collisions and deviations on a per vocabulary term basis.
- Interoperability – is achieved with the M-M and H-M use of singular terms across the enterprise and/or the ability to mediate vocabulary labels based on the globally unique vocabulary tags.

The Linked and Security elements of VAULTIS are based on application of CCV/LCV and not directly attributable to the model.

As the CCV/LCV model is implemented, the creation of data profiles (xBOMs) can be populated with known Data Tags and Labels and improve the ability to understand specific data characteristics and constructed sets of information by machines and humans alike. This has a direct impact on shifting the focus for a data consumer from first evaluating the data on receipt, to immediately applying the data to mission purpose.